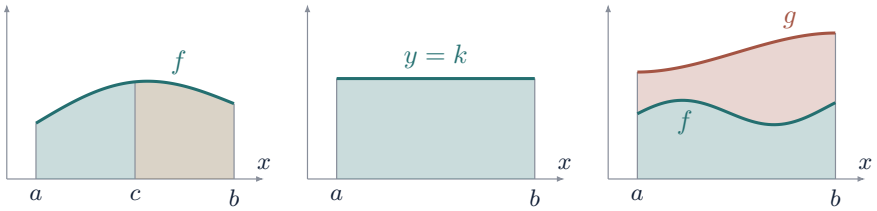
Mathematics Winter 2027

Calculus II

MATH 1952 · CRN 1540 · Undergraduate

Calculus is the study of functions on the continuum, their variations, their summations, and other such notions. The fundamental tool of calculus is the idea of limits and the rigorous notion of approximation. It is the foundation of all modern science, the language which we use to model the world, and the greatest discovery by the greatest minds. This course introduces the integral of continuous functions on compact intervals and the Fundamental Theorem of Calculus, with applications to the computation of lengths, areas, volumes and averages. The integral is the continuum summation counterpart to the infinitesimal difference, or derivative.



Chasles relation
$\int_a^b f = \int_a^c f + \int_c^b f$

Normalization
$\int_a^b k = k(b-a)$

Comparison
$\int_a^b f \leq \int_a^b g$

Fundamental theorem of calculus

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

Topics

- The integral of continuous functions
- The Fundamental Theorem of Calculus
- Antiderivatives
- Riemann sums
- Linearity of the integral, substitution and integration by parts
- Trigonometric and inverse trigonometric substitutions
- Applications to curve lengths, areas and volumes

Prerequisites

Calculus I

Instructor

Prof. Frédéric Latrémolière

Meetings

Tuesdays & Thursdays
12:00–1:50 p.m.
CMK 101

September 8–November 16

TA session

Fridays
12:00–12:50 p.m.
CMK 101